

Ioannis Koutalios

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Experience	Sensorfact	<i>Utrecht, Netherlands</i>
	Data Scientist	Jul 2026 – Present
	Junior Data Scientist	Jan 2026 – Jun 2026
	Junior Data Analyst	Mar 2025 – Dec 2025
	Leiden University	<i>Leiden, Netherlands</i>
	Graduate Student Researcher	Mar 2022 – Jun 2024
	<i>Teaching Assistant, "Reinforcement Learning" (M.Sc.)</i>	Jan – Jun 2024
	<i>Teaching Assistant, "Science and the Public" (M.Sc.)</i>	Mar 2023 – Jul 2024
	<i>Teaching Assistant, "Astronomy Lab and Observing Project" (B.Sc.)</i>	Mar 2024
	Aristotle University of Thessaloniki	<i>Thessaloniki, Greece</i>
Education	Undergraduate Student Researcher	Sep 2019 – Nov 2020
	<i>Gravitational Waves Group</i>	
	<i>Grading Assistant, "Astronomy and Astrophysics"</i>	Mar 2018 – Jun 2020
	M.Sc., Astronomy and Data Science	Sep 2024
	Leiden University	<i>Leiden, Netherlands</i>
Publications	B.Sc., Physics	Nov 2020
	Aristotle University of Thessaloniki	<i>Thessaloniki, Greece</i>
	The carbon isotope ratio of β Pic b with high-resolution spectroscopy	
	González Picos, D., Snellen, I. A. G., Landman, R., de Regt, S., Grasser, N., Birkby, J. L., Stolker, T., Koutalios, I. & Kenworthy, M. A.	
	<i>Astronomy & Astrophysics</i>	Jul 2026
	doi: 10.1051/0004-6361/202660276	
	Upper limits on exosatellites around β Pictoris b	
	Kenworthy, M. A., Landman, R., Vanderburg, A., Rodriguez, J. E., Birkby, J. L., Macias, I., González Picos, D., Jenkins, S. A., Kleisioti, E., Stolker, T. & Koutalios, I.	
	<i>Monthly Notices of the Royal Astronomical Society</i>	Jul 2026
	doi: 10.1093/mnras/stag1060	
	The resummation model in FLAMINGO: precisely predicting matter power suppression from observed halo baryon fractions	
	van Daalen, M. P., Koutalios, I. , Broxterman, J. C., Wolfs, B. J. H., Helly, J. C., Schaller, M. & Schaye, J.	
	<i>Monthly Notices of the Royal Astronomical Society</i>	Nov 2025
	doi: 10.1093/mnras/staf2086	
	Reinforcement learning for sensorless astronomical imaging	
	Morisse, J., Koutalios, I. , Landman, R. & Moerland, T.	
	<i>36th Benelux Conference on Artificial Intelligence (BNAIC/BeNeLearn)</i> , extended abstract	Nov 2024
	pdf: bnaic2024 code: johnkou97/AdaptiveOptics	

Research	Predicting binary neutron star postmerger spectra using artificial neural networks Pesios, D., Koutalios, I., Kugiumtzis, D. & Stergioulas, N. <i>Physical Review D</i> doi: 10.1103/PhysRevD.110.063008 Applying a cross-power model for power suppression in FLAMINGO <i>M.Sc. Thesis, sup. Dr. Marcel van Daalen</i>	Sep 2024 Sep 2023 – Jun 2024
	Model-Free Reinforcement Learning for Sensorless Adaptive Optics <i>Research Project, sup. Dr. R. Landman, Dr. T. Moerland, Dr. A. Plaat</i>	Oct 2023 – Feb 2024
	Finding the best exoplanets to search for exomoons by radial velocity <i>M.Sc. Thesis, sup. Dr. Matthew Kenworthy</i>	Mar 2022 – Dec 2022
	Spectral Classification of Neutron Star Post-Merger Gravitational Wave Emission <i>B.Sc. Thesis, sup. Dr. Nikolaos Stergioulas</i> <i>Gravitational Waves Group, Aristotle University of Thessaloniki</i>	Sep 2019 – Sep 2020
	BNAIC/BeNeLearn 2024: Joint International Scientific Conferences on AI Poster: Model-Free RL for Sensorless Adaptive Optics	Utrecht, Nov 2024
Conferences	79th Dutch Astronomers' Conference Poster: Applying a cross-power model for power suppression in FLAMINGO	Egmond aan Zee, May 2024
Volunteer	Instructor for the Preparation of the Greek Team for the International Olympiad on Astronomy and Astrophysics	2016 – 2023
	Public Outreach Guide Astronomical Observatory of the Aristotle University of Thessaloniki	2018 – 2021
Distinctions	Scholarship from Cosmote for undergraduate studies	2015 – 2019
	International Olympiad on Astronomy and Astrophysics Honorable Mention (<i>Indonesia</i>)	2015
	Participation (<i>Romania</i>)	2014
Skills	Core Programming & Cloud Python, TypeScript, SQL, AWS, Git (GitHub & GitLab), Linux, $\LaTeX$	
	Data Science & Machine Learning PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, SciPy	
	Infrastructure & Tools Prefect, Grafana, Bash, SSH	
Languages	Greek (Native), English (C2), German (B1), Dutch (Beginner)	